Muhammet Celal Gokalp

ADAS & Autonomous Systems Engineer | Systems, Validation & Customer Integration

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PROFESSIONAL SUMMARY

Systems and application engineer with nearly 8 years in ADAS and automated driving. I worked on-site at Jaguar Land Rover as an ADAS feature owner, and since 2021 I have worked on an L4 automated driving stack at FiveAI / Bosch. My work covers the full cycle: turning customer needs and regulation into requirements and system states, defining interfaces, and proving the result through simulation, HiL / ViL rigs and vehicle testing. I work well between customers, engineering teams and suppliers, and I use AI tools to build practical engineering tools and help colleagues adopt them.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – 11/2026

FiveAI / Bosch — Cambridge, United Kingdom

- **Scenario-based verification:** Built 120+ parameterised OpenSCENARIO 2 scenarios generating thousands of test cases for an L4 automated driving stack, covering UN R79 automatic lane change (ADA project) and Euro NCAP junction tests.
- **Acceptance criteria & analysis:** Defined pass/fail and KPI criteria, such as comfort limits on acceleration and deceleration, and analysed results across 1,000+ hours of real-world and simulation data.
- **AI-assisted engineering:** Built LLM agent skills with GitHub Copilot and Copilot CLI that turn regulation and Euro NCAP rules into scenarios. Adopted project-wide, they increased scenario output by an order of magnitude. Coached colleagues on using AI tools and checking their output.
- **Integration & support:** On the ADA project, a partnership with Volkswagen Group where FiveAI supplied the simulation tooling, trained partner engineers to use our toolchain. This grew into the Simulation EcoSystem solutions team, where I was a founding member. Triaged integration issues across the stack, tools, maps (SDB / XODR) and simulation, working with several engineering teams to find root causes.
- **Tools & pipelines:** Worked in the project monorepo and its dev container, ran simulations through the CI test pipeline, and built Python tools and data pipelines that send results to dashboards.

ADAS Systems Engineer (Feature Owner)

11/2018 – 10/2021

AVL Powertrain UK — Coventry, United Kingdom

- **Customer-embedded feature owner:** Based at Jaguar Land Rover as owner of Rear Cross Traffic Alert, Blind Spot Monitor and Rear Pre-Crash (corner radar features). Main technical link between the OEM, AVL and the sensor supplier.
- **Requirements & interfaces:** Wrote use cases, system state diagrams and requirements, broken down from system to component level in DOORS and MKS Integrity (ASPICE V-model). Defined and reviewed input and output signal lists for each module, mapped to the state machines.
- **Validation & diagnosis:** Ran verification on HiL and ViL rigs and on the proving ground, analysing CAN data with Vector CANoe and CANape. For example, I traced a late alert at boundary conditions to sensor field of view and agreed a fix with the supplier while keeping false alerts low.
- **Field diagnostics:** At the LEVC (London Taxi) factory, inspected returned faulty e-machines: multimeter checks, downloading data from proprietary loggers and sending it to developers and suppliers for root cause analysis.
- **Simulation correlation:** Calibrated a Simulink / IPG CarMaker vehicle model against real vehicle data, including parameter sweeps on steering torque PID gains, to compare vehicle, dyno and simulation testing.
- **Feature design & safety:** Wrote requirements for Adaptive Cruise Control, Lane Keep Assist, Highway Assist and Lane Change Assist. Wrote ISO 26262 functional safety requirements for a lane change feature; trained in ISO 26262 and SOTIF.

Mechanical Engineering Intern

05/2014 – 08/2014

TEMSA Global — Adana, Türkiye

- Calculated power and torque requirements for a new electric bus powertrain using MATLAB / Simulink.

Part-Time Teacher and Program Planner

12/2013 – 01/2017

Space Camp Turkey — Izmir, Türkiye

- Taught space technology and science to students aged 9 to 16, and planned programme schedules for groups and teachers.

EDUCATION

MSc in Sustainable Automotive Engineering

10/2017 – 09/2018

Warwick Manufacturing Group, Coventry, UK

- **Grade:** 2:1 (Merit). Dissertation: experimental thermal modelling of an electric motor stator (MATLAB / Simulink, MotorCAD, Portunus).

BSc in Mechanical Engineering

08/2011 – 02/2017

Ege University, Izmir, Türkiye

- Dissertation: designed, built and controlled a 6-DOF Stewart motion platform (state-space modelling in MATLAB, control on Arduino).

SKILLS

Systems Engineering	Requirements and decomposition, system states, interface and signal definition, traceability, V-model / ASPICE, DOORS, MKS Integrity
Validation	Scenario-based testing, SIL, HiL / ViL, proving ground testing, KPI and data analysis, CI regression
Safety & Regulation	ISO 26262 (requirements written, trained), SOTIF / ISO 21448, UN R79, Euro NCAP
Software & Tools	Python, C++ (reading), OpenSCENARIO 2, Git, dev containers, MATLAB / Simulink, IPG CarMaker, Vector CANoe / CANape, Jira / Confluence, GitHub Copilot
Other	British national, full UK driving licence. English (C2), Turkish (Native)